

Agent Specifications

Common Agent Structure

Every agent is a standalone Python application with:

Component	Purpose
server.py	FastAPI HTTP server -- interface to orchestrator
brain.py	Multi-step reasoning engine (analyze -> reflect ->)
analyst.py	Statistical tools (pandas/numpy) for data analysis
memory.py	SQLite database for decisions, reasoning, observati
tools.py	Python functions the reasoning engine can invoke
config.yaml	Role, model, fingerprint, orchestrator URL
prompts/	Role-specific prompt templates for each reasoning s
data/shared/	Read-only mirror: world docs + past simulation data
data/private/	Private database, grows over the run

Common Interface

```
# What the orchestrator calls (HTTP)
POST /turn      <- context JSON      -> decision JSON
POST /sync      <- shared data        -> ack
POST /reset     <- new run config     -> ack
GET  /health    -> {"status": "ok", "agent_id": "firm_0"}
GET  /memory/summary -> condensed memory for diagnostics
```

Common Reasoning Pipeline

Every agent type follows the same 6-step pipeline per turn. The CONTENT differs by agent type, but the STRUCTURE is identical:

```
OBSERVE -> ANALYZE -> REFLECT -> STRATEGIZE -> DECIDE -> STORE
(parse)   (stats)   (LLM #1)   (LLM #2)   (LLM #3)   (DB write)
```

Fingerprints (Personality Profiles)

Each agent receives a randomly drawn personality at run start. The fingerprint is:

- Fixed for the run (deterministic from run seed + agent index)
- Included in the system prompt
- Designed to create heterogeneity across agents of the same type

Fingerprint dimensions (all agents):

- risk_appetite: 0.0 (very conservative) to 1.0 (very aggressive)
- time_horizon: "short" (2-3yr), "medium" (5-7yr), "long" (10+yr)
- style: role-specific (see below)
- narrative_personality: a paragraph describing the agent's philosophy

Agent 1: Environment

Identity

- agent_id: env_0
- agent_type: environment
- 1 instance, typically on the strongest machine / largest model

Role

The "game master" -- takes firm actions as input, produces market outcomes. Maintains narrative continuity of the simulated world.

Fingerprint Dimensions

```
style: "balanced"           # balanced, volatile, conservative, disruption-prone
macro_volatility: 0.6       # tendency to generate large vs. small shocks
event_frequency: 0.3       # tendency to generate special events
narrative_style: "analytical" # analytical, dramatic, terse
```

Reasoning Pipeline (per turn)

Step 1 - OBSERVE: Parse all firm actions, current macro state, active events.

Step 2 - ANALYZE (statistical):

- Compute multinomial logit demand baseline from prices/qualities/brands
- Check active event durations (which expire this quarter?)
- Compare firm actions to historical patterns from past simulations
- Compute aggregate statistics: industry revenue, average price, capacity utilization

Step 3 - REFLECT (LLM call #1): Prompt: "Given these firm actions and market conditions, what is happening in this market this quarter? Are firms competing on price or quality? Is the market growing or stagnating? Are any firms behaving unusually?"

Step 4 - STRATEGIZE (LLM call #2): Prompt: "Determine market outcomes. Consider: (a) How much total demand materializes given awareness, prices, and product quality? (b) How is demand allocated across firms? (c) Do any R&D programs succeed this quarter? (d) Do any special events occur? Use the baseline demand model as a reference but you may deviate if you have a good reason."

Step 5 - DECIDE (LLM call #3): Prompt: "Produce the final outcome JSON. Also write a 2-3 paragraph narrative explaining what happened this quarter and why. This narrative will be shared with all agents."

Step 6 - STORE: Save the full reasoning trace + outcomes.

Output Schema

```
{
  "total_demand": 88500,
  "demand_rationale": "...",
  "firm_outcomes": [
    {"firm_id": "firm_0", "units_sold": 19200, "market_share": 0.217}
  ],
  "rd_outcomes": [
    {"firm_id": "firm_0", "product_advance": false, "process_cogs_reduction": 0.01, "delivery_advance": false}
  ],
  "events": [
    {"type": "none|safety_scandal|regulatory_change|breakthrough|supply_disruption|macro_shock",
```

```

    "description": "...", "affected_firms": [], "duration_quarters": 0, "demand_impact": 0.0}
  ],
  "macro_update": {"awareness_change": 0.02, "regulatory_mood": "stable"},
  "narrative": "Q2 2031 saw continued growth in SRT adoption..."
}

```

Guardrails (enforced by orchestrator)

- total_demand in [0.3x, 3.0x] of multinomial logit baseline
- sum(units_sold) == total_demand
- units_sold[i] <= production[i] for each firm
- No firm market share > 60%
- R&D advances respect cumulative thresholds from world docs
- Event probabilities respect base rates (no scandal every quarter)

Agent 2: Firm (x5 instances)

Identity

- agent_id: firm_0 through firm_4
- agent_type: firm
- 5 instances, each with unique fingerprint

Role

Runs a pharmaceutical company. Makes all operational and financing decisions. Builds and defends competitive position over 20 years.

Fingerprint Dimensions

```

style: "growth-focused"           # growth, profitability, rd-intensive, marketing-driven, conservative, b
risk_appetite: 0.72
time_horizon: "long"
innovation_priority: "efficacy"   # efficacy, safety, convenience, cost
financing_preference: "equity"    # equity, debt, balanced
narrative_personality: "We are a science-first company..."

```

What Firms Know (depends on information regime -- see doc 06)

Always available (private):

- Own complete balance sheet, income statement, cash flows
- Own internal stocks (capability, brand, capacity, unit cost)
- Own R&D progress (cumulative spending by program, generation status)
- Own decision history and reasoning traces (from memory.db)

Available under baseline regime (public):

- Competitors' published financial statements (level of detail depends on measurement regime)
- Equity prices for all firms

- Macro state (risk-free rate, market growth indicators)
- Environment narrative (qualitative description of market events)
- Shared past simulation database (Compustat panels from prior runs)

Never available:

- Competitors' private internal stocks (capability, brand levels)
- Competitors' R&D allocation across programs
- Competitors' unit costs
- Financial institutions' internal portfolio decisions about OTHER firms
- (Unless information regime is set to "full_transparency" for research)

Reasoning Pipeline (per turn)

Step 2 - ANALYZE (statistical, unique to firms):

- Margin analysis: gross, operating, net margin trends over last 4-8 quarters
- Revenue decomposition: price effect vs. volume effect vs. market growth
- R&D efficiency: capability gain per dollar invested, by program
- Cash runway: quarters of runway at current burn rate
- Competitor analysis: price trends, share trends, inferred strategies
- Historical comps: "firms like me in past simulations did X and got Y"
- Capacity utilization: production vs. capacity, and demand vs. production

Step 3 - REFLECT (LLM call #1): "Here is your financial summary, competitive position, and historical comparisons. What is your assessment of the situation? What are your strengths, weaknesses, and the key strategic questions for this quarter?"

Step 4 - STRATEGIZE (LLM call #2): "Given your assessment, generate 2-3 strategic options. For each, specify the key decision variables (price, R&D, capex, marketing) and the expected tradeoffs. Consider your cash position -- you have \$X in cash and \$Y in available credit. Your capacity is K units. Do NOT plan to spend more than you can afford."

Step 5 - DECIDE (LLM call #3): "Choose your plan for this quarter. Output the decision JSON. Justify each major number."

Output Schema

```
{
  "price": 92000,
  "production": 220,
  "capex": 20000000,
  "rd_spend": 30000000,
  "rd_allocation": {"product": 0.55, "process": 0.25, "delivery": 0.20},
  "sga_spend": 15000000,
  "equity_issuance_request": 0,
  "debt_request": 0,
  "dividends": 0,
  "buybacks": 0,
  "reasoning": "Lowering price slightly to capture volume. Heavy R&D push..."
}
```

Agent 3: Investment Bank (1 instance)

Identity

- agent_id: ibank_0
- agent_type: investment_bank

Role (Revised -- equity pricing moved to Equity Market agent)

- Advisory and structuring: structures IPOs, secondary offerings, M&A deals
- Does NOT price equity (that is the Equity Market agent's role)
- Acts as intermediary: firms request capital -> IBank structures -> Equity Market decides
- If ma_enabled: advises on acquisition strategy, values targets, structures deals
- Publishes equity research and recommendations (advisory, not price-setting)

Fingerprint Dimensions

```
style: "relationship-advisory"    # relationship, aggressive-dealmaker, conservative, balanced
deal_appetite: 0.55
advisory_approach: "fundamental valuation with strategic premium"
fee_sensitivity: 0.3              # willingness to discount fees for repeat clients
```

What the Investment Bank Does Each Quarter

- IPO/secondary structuring (if any firm requests equity):
 - Reviews firm's request and financials
 - Recommends share count, price range, offering structure
 - Passes structured offering to Equity Market for subscription decision
 - Earns underwriting fee (5% of proceeds if successful)
- M&A advisory (if ma_enabled and any firm proposes acquisition):
 - Values the target firm
 - Structures the deal (cash/stock mix, financing plan)
 - Recommends to both parties
 - Earns advisory fee (2% of deal value)
- Equity research (every quarter):
 - Publishes ratings (overweight/neutral/underweight) and target prices
 - These are OPINIONS, not binding prices
 - Firms and other agents see the research

Output Schema

```
{
  "equity_research": [
    {
      "firm_id": "firm_0",
      "rating": "overweight",
      "target_price_4q": 42.00,
      "reasoning": "Strong R&D pipeline approaching Gen 2..."
    }
  ],
  "ipo_structuring": [
    {
```

```

    "firm_id": "firm_3",
    "recommended_price_range": [14.00, 17.00],
    "recommended_shares": 12000000,
    "offering_structure": "firm_commitment",
    "reasoning": "Conservative range for new entrant..."
  }
],
"ma_advisory": [],
"commentary": "SRT sector showing diverging fundamentals..."
}

```

Agent 3b: Equity Market (1 instance) -- NEW

Identity

- agent_id: eqmkt_0
- agent_type: equity_market

Role

The Equity Market represents the aggregate behavior of equity investors -- private equity and venture capital when firms are private, public stock market investors when firms are public.

This is the agent that actually prices equity and decides whether to invest.

Mode-Dependent Behavior

Private mode (firm is private):

- Acts as a PE/VC consortium evaluating funding rounds
- Has MORE information than public mode (due diligence access)
- Applies an illiquidity premium (higher required return)
- Can impose governance terms (board seats, liquidation preferences)
- Sets "last-round valuation" each quarter (portfolio mark-to-market)
- Decides: fund this round? At what valuation? How much capital?

Public mode (firm is public):

- Acts as the aggregated stock market
- Prices equity based on DCF + multiples + sentiment + momentum
- Decides whether to subscribe to secondary offerings (and at what discount)
- Has LESS information than private mode (only public financials)
- No governance terms beyond shareholder voting
- Sets market price each quarter

IPO transition: When a private firm IPOs, the Equity Market shifts from PE to public mode for that firm. The IBank structures the offering; the Equity Market sets the market-clearing price and subscribes.

Fingerprint Dimensions

```

style: "fundamental-value"      # fundamental-value, growth-investor, momentum-trader, contrarian
risk_appetite: 0.60
sentiment_sensitivity: 0.4      # how much macro mood affects pricing

```

```

illiquidity_premium: 0.05      # extra return demanded for private equity (quarterly)
portfolio_concentration_limit: 0.30 # max % of portfolio in one firm

```

Valuation Reasoning (see doc 07 for full framework)

The Equity Market agent uses the multi-step reasoning pipeline to price firms:

- Analyze financials (statistical tools): revenue trajectory, margins, cash, debt, R&D progress, market share trends
- Project scenarios (LLM): base case (50%), upside (25%), downside (25%) with revenue, margin, and FCF projections

- Compute value (statistical): probability-weighted DCF
- Cross-check (LLM): revenue multiples, past simulation comps, reasonableness
- Set price (LLM): final price with justification and confidence level

What the Equity Market Knows

Information	Private Mode	Public Mode
Financial statements	Full (due diligence)	Published only (per info regime)
R&D pipeline detail	Full (board access)	Summary only
Internal capability/brand	Yes (board reporting)	No
Unit cost detail	Yes	No
Customer satisfaction	Yes	Public scores only
Competitor private info	No (only own portfolio)	No

Output Schema (Public Mode)

```

{
  "equity_prices": [
    {
      "firm_id": "firm_0",
      "price_per_share": 34.00,
      "valuation_method": "DCF",
      "key_assumptions": {
        "revenue_growth_5yr": 0.35,
        "terminal_margin": 0.30,
        "discount_rate": 0.15,
        "terminal_multiple": 18
      },
      "base_case_value": 38.00,
      "downside_value": 22.00,
      "confidence": "medium",
      "reasoning": "Strong R&D pipeline approaching Gen 2..."
    }
  ],
  "subscription_decisions": [
    {
      "firm_id": "firm_3",
      "offering_type": "ipo",
      "subscribe": true,
      "shares_subscribed": 12000000,

```

```

    "price_accepted": 15.50,
    "reasoning": "Fair pricing given early stage. Willing to take full allocation."
  }
],
"market_sentiment": "cautiously_bullish"
}

```

Output Schema (Private Mode)

```

{
  "private_valuations": [
    {
      "firm_id": "firm_0",
      "post_money_valuation": 800000000,
      "implied_price_per_share": 16.00,
      "valuation_method": "comparable_transactions",
      "key_assumptions": {
        "revenue_multiple": 12,
        "pipeline_premium": 1.3,
        "illiquidity_discount": 0.25
      },
      "reasoning": "Comparable to Series C biotech valuations..."
    }
  ],
  "funding_decisions": [
    {
      "firm_id": "firm_2",
      "fund": true,
      "amount": 75000000,
      "valuation_pre_money": 400000000,
      "equity_stake_pct": 15.8,
      "terms": {
        "liquidation_preference": "1x_non_participating",
        "board_seat": true,
        "anti_dilution": "weighted_average"
      },
      "reasoning": "Early-stage bet on promising R&D team."
    }
  ]
}

```

Agent 4: Commercial Bank (1 instance)

Identity

- agent_id: cbank_0
- agent_type: commercial_bank

Role

- Provides revolving credit facilities (working capital)

- Manages credit risk through commitment sizing and pricing
- Can be forced into losses if firms default (see doc 07)
- Can itself become distressed if cumulative losses exceed capital

Fingerprint Dimensions

```
style: "relationship-lender"    # aggressive, conservative, relationship, opportunistic
risk_appetite: 0.45
loss_tolerance: 0.05           # max tolerable loss rate
concentration_limit: 0.30      # max % of portfolio to single firm
```

Bank Capital and Failure (see doc 07 for full details)

The commercial bank starts with a capital base (set in config, e.g., \$2B). Losses from defaults erode capital. If capital falls below a regulatory minimum (e.g., 8% of outstanding commitments), the bank becomes constrained:

- Must reduce commitments (cannot extend new credit)
- Must raise interest rates to rebuild capital
- If capital reaches zero, the bank fails and is replaced

What the Commercial Bank Knows

- All firms' published financial statements (per measurement regime)
- Its own loan portfolio (outstanding balances, rates, covenants, loss history)
- Macro state
- Environment narrative
- Shared past simulation database
- Default recovery rates from past simulations

Reasoning Pipeline (unique elements)

ANALYZE step includes:

- Portfolio concentration analysis (exposure per firm as % of capital)
- Coverage ratio analysis per firm (EBITDA / interest, cash / debt)
- Liquidity analysis per firm (cash runway, revolver utilization)
- Historical default prediction (firms with similar metrics in past sims)
- Own capital adequacy check (current capital / total commitments)

Output Schema

```
{
  "revolver_terms": [
    {
      "firm_id": "firm_0",
      "commitment": 75000000,
      "rate_quarterly": 0.020,
      "maturity_quarters": 4,
      "covenants": {
        "min_cash": 20000000,
        "max_leverage": 3.0,
        "revenue_decline_trigger": -0.30
      }
    },
  ],
}
```

```

    "reasoning": "Maintaining $75M facility. Firm has strong cash position..."
  }
],
"own_capital_status": {
  "capital": 1850000000,
  "total_commitments": 350000000,
  "capital_ratio": 0.189,
  "status": "well_capitalized"
}
}

```

Agent 5: Credit Fund (1 instance)

Identity

- agent_id: cfund_0
- agent_type: credit_fund

Role

- Provides term debt (longer-duration financing for growth, capex, acquisitions)
- Prices credit risk through interest rates and structures
- Subject to same capital/failure dynamics as commercial bank (see doc 07)

Fingerprint Dimensions

```

style: "growth-capital"          # conservative, growth-capital, distressed, yield-focused
target_return: 0.10              # annual target return on deployed capital
max_single_exposure: 200000000
preferred_maturity_quarters: 12
capital_base: 3000000000         # starting capital

```

What the Credit Fund Knows

Same as commercial bank, plus:

- Term debt outstanding per firm (balances, rates, maturities)
- Amortization schedules
- Seniority structure (its term debt vs. the commercial bank's revolver)

Output Schema

```

{
  "term_debt_terms": [
    {
      "firm_id": "firm_0",
      "max_new_issuance": 100000000,
      "rate_quarterly": 0.030,
      "maturity_quarters": 12,
      "seniority": "senior_secured",
      "amortization": "bullet",
      "covenants": {

```

```

    "max_total_leverage": 4.0,
    "min_interest_coverage": 2.0
  },
  "reasoning": "Offering growth capital to fund capacity expansion..."
}
],
"own_capital_status": {
  "capital": 2900000000,
  "total_deployed": 280000000,
  "unrealized_losses": 0,
  "status": "healthy"
}
}

```

Agent Lifecycle Events

Run Start

- Orchestrator generates fingerprints from seeded RNG
- Orchestrator sends POST /reset to each agent with:
 - config.yaml content (agent_id, fingerprint, regime settings)
 - Shared data (world docs, past simulation summaries)
- Each agent initializes its database and builds its system prompt

Normal Quarter

- Orchestrator sends POST /turn with phase-appropriate context
- Agent runs reasoning pipeline, returns decision
- Orchestrator sends POST /sync with updated shared data after quarter ends

Firm Default

- Orchestrator notifies the defaulting firm agent: "You have defaulted. Produce a final reflection on what went wrong."
- Orchestrator notifies financial institutions: "Firm X has defaulted. Your exposures are [amounts]. Recovery is [amount]."
- The firm agent for that incarnation receives POST /reset with new incarnation config (new fingerprint, clean slate, but slot history included)
- Financial institutions update their portfolios

Financial Institution Distress

- If a bank/fund's capital falls below threshold, the orchestrator:
- Notifies the institution: "Your capital ratio is [X%], below [threshold].

You must reduce exposures."

- Notifies firms: "Your [revolver/term debt] provider is distressed. Terms may tighten."
- If capital reaches zero: institution is replaced (new fingerprint, fresh capital)

- See doc 07 for full failure mechanics

Death-Spiral Prevention (Firm Slots)

- 3 consecutive Q1 defaults: slot paused 1 quarter
- 6 consecutive failures: slot frozen permanently
- Financial institutions see slot default history in their context

Run End

- Orchestrator sends POST /turn with phase: "debrief" to every agent
- Each agent produces a final reflection (stored in debrief.csv)
- Agents' private databases are archived for potential post-run analysis